PLAYER-BASED CRICKET VIDEOS SEGMENTATION SYSTEM

25-26J-148-DS

Project Proposal Report

R M N D Rathnayake

B.Sc. (Hons) Degree in Information Technology Specialized in Data Science

Department of Information Technology

Sri Lanka Institute of Information Technology
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DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

Video analysis in cricket has become a critical tool for performance evaluation, coaching, and viewer engagement. However, traditional methods rely heavily on manual tagging and review, making the process inefficient and difficult to scale. This research introduces an automated deep learning–based approach for cricket video segmentation by player identity. The proposed system integrates Optical Character Recognition (OCR) of scoreboard frames with speech-to-text conversion of commentary audio, followed by Natural Language Processing (NLP) to identify player mentions. By synchronizing extracted text from the scoreboard and audio streams, the system automatically segment video based on player without manual intervention. This eliminates the need for labor-intensive labeling while enabling efficient indexing of full-length cricket matches. The novelty of this component lies in its multimodal fusion of OCR and audio analysis, providing a scalable solution for automated player-based video segmentation in cricket. The outcomes are expected to significantly enhance the accessibility of targeted match highlights for coaches, analysts, and fans.

Keywords - Cricket video analysis, Deep learning, video segmentation ,Player identification, Optical Character Recognition (OCR),Scoreboard ,Speech-to-text conversion, Natural Language Processing (NLP),Multimodal fusion, Commentary analysis, Sports analytics, evaluation, Scalable video indexing, Coaching and training support

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1. INTRODUCTION

Cricket, a globally celebrated sport, generates extensive video content from matches, yet fans, coaches, and analysts face significant challenges in accessing specific player performances due to the time-consuming task of manually navigating full-length videos to locate moments featuring favorite players [1], [2]. This inefficiency underscores the need for automated solutions to segment cricket videos by player identity, enhancing accessibility for performance analysis, coaching, and viewer engagement [3], [4]. Research in automated sports video analysis has progressed significantly, evolving from early rule-based methods for event detection, such as boundary or wicket identification using color histograms and edge detection, to sophisticated deep learning approaches for highlight generation and action recognition [1], [5], [6]. Recent studies have explored convolutional neural networks (CNNs) for classifying cricket shots and wickets, achieving accuracies above 85% in event detection, and have developed datasets for cricket-specific tasks like ball tracking and delivery analysis [3], [7], [4]. However, these efforts primarily focus on event-driven summarization rather than player-specific segmentation, leaving a gap in targeted player-focused video analysis [1], [4].

To address this project's objective of automatic cricket video segmentation by player identity, expertise in several domains is essential. Key techniques include computer vision for object detection and tracking to identify players in video frames, optical character recognition (OCR) to extract player names from dynamic scoreboards, and audio processing for speech-to-text (STT) analysis of commentary to detect player mentions [1], [3], [4]. Specifically, OCR models like PaddleOCR or EasyOCR are needed to handle scoreboard text variations under diverse lighting and motion conditions [8]. STT models, such as OpenAI Whisper or Wav2Vec2, are critical for transcribing commentary in noisy environments, followed by natural language processing (NLP) using tools like spaCy for named entity recognition to identify player mentions [9]. Multimodal fusion, leveraging transformer-based architectures or LSTM for temporal alignment, is necessary to

synchronize video, audio, and text data for accurate segmentation [10]. Additionally, proficiency in deep learning frameworks (e.g., PyTorch, TensorFlow) and cricket domain knowledge ensures contextually relevant outputs [4], [2].

The state of the art in sports video analysis has advanced toward multimodal deep learning systems, achieving over 90% accuracy in action recognition and player tracking in sports like soccer and basketball [11], [12]. In cricket, recent frameworks utilize YOLO-based models for real-time ball tracking and MobileNet for lightweight shot classification, with datasets like those from IEEE DataPort enabling fine-grained analysis [3], [7]. Multimodal approaches integrating video and audio, such as those using ViViT or CLIP for semantic understanding, represent cutting-edge progress, though primarily applied to highlight generation rather than player segmentation [10], [13]. Commercial tools like Veo's Player Spotlight demonstrate automated tracking in team sports, but cricket-specific applications remain limited [14]. Previous attempts to address player-focused video access have relied on event-based segmentation, such as detecting wickets or shots using CNNs or SVMs, often requiring manual post-processing to filter player-specific moments [1], [3], [6]. Some systems use audio excitement cues or replay detection for highlights but lack integration of scoreboard OCR and commentary STT for precise player identification [4], [5]. Unlike these approaches, this project proposes an end-to-end multimodal system combining OCR-extracted scoreboard data, STT-processed commentary, and video-based player detection to automatically segment videos by player identity without manual tagging. This builds on prior work by enhancing scalability for full-match analysis and introducing synchronized multimodal processing for real-time player-specific filtering, offering a novel solution for fans and analysts to access favorite player moments efficiently [1], [4], [10].

1.1. Background and Literature Survey

Cricket, with its global fanbase exceeding billions, generates vast amounts of video content, yet the manual effort required to locate specific player performances remains a significant barrier for fans, coaches, and analysts [1], [2]. The primary challenge addressed in this project is the inability of users to efficiently access segments of cricket videos featuring their favorite players without navigating entire match recordings [3], [4]. This inefficiency hampers strategic analysis, coaching, and viewer engagement, necessitating automated systems for player-specific video segmentation [7], [6]. Over the past two decades, research in sports video analysis has transitioned from traditional signal processing to advanced deep learning techniques, driven by the need for scalable and accurate automation in processing unstructured multimedia data [5], [11].

Early work in cricket video analysis focused on rule-based methods for event detection, such as identifying boundaries or wickets using low-level features like color histograms, edge detection, and audio cues from crowd excitement [6], [15]. For instance, Kolekar et al. utilized audio-visual cues to segment cricket videos into shots, achieving 80% accuracy in detecting key events like sixes or fours [15]. However, these methods lacked specificity for player identification and relied heavily on manual post-processing [1]. The advent of deep learning marked a shift toward more robust solutions, with convolutional neural networks (CNNs) like VGG and ResNet applied to classify cricket shots and wickets, achieving accuracies of 85–90% on small-scale datasets [7], [16]. Raza et al. proposed a player-specific framework for highlight generation using CNN-based action recognition, integrating visual features to summarize key moments, though it did not address full-match player segmentation [1].

Recent advancements have leveraged multimodal approaches to enhance sports video analysis. Ansari et al. developed a dataset and deep learning pipeline for front-pitch view shot extraction and ball tracking in cricket, employing YOLO and MobileNet to achieve 92% accuracy in delivery detection [3]. Similarly, Reddy utilized a superior CNN

architecture for bowling delivery classification, reporting 90% accuracy in distinguishing spin and fast deliveries [7]. These studies, while effective for event-specific tasks, do not fully address player-centric segmentation, particularly for full-length matches [4]. Multimodal systems integrating audio and video have emerged, with frameworks like SmartCrick combining excitement detection from commentary and visual cues for highlight generation, yet they lack precise player identification via scoreboard or commentary analysis [4], [17].

In the broader sports domain, state-of-the-art systems have advanced player tracking and identification. For example, Giancola and Ghanem's FineSports dataset supports fine-grained action recognition in team sports, using 3D CNNs and pose estimation to achieve 91% accuracy in multi-player tracking [11]. PlayerTV and similar tools in soccer utilize YOLO-based detection and re-identification for real-time player tracking, with mean average precision (mAP) exceeding 90% [12], [18]. Multimodal transformers, such as ViViT and CLIP, have been applied to integrate video, audio, and text for semantic analysis, achieving 93% accuracy in action localization and video summarization [10], [19]. In cricket, Hawk-Eye's ball-tracking systems and AI-driven coaching tools integrate visual data with contextual metadata, but they focus on ball trajectory and game statistics rather than player-specific video segmentation [20].

For player identification, optical character recognition (OCR) and speech-to-text (STT) techniques are critical. PaddleOCR and EasyOCR have been employed for dynamic text extraction in sports videos, handling challenges like motion blur and font variations with 95% word accuracy on benchmark datasets like ICDAR [8], [21]. STT models like OpenAI Whisper and Wav2Vec2 excel in noisy environments, achieving word error rates (WER) below 10% for commentary transcription, with subsequent NLP using BERT or spaCy for named entity recognition to extract player mentions [9], [13], [22]. Multimodal fusion, using transformers or LSTM-based architectures, synchronizes these inputs for temporal alignment, as demonstrated in video summarization tasks with 90% segmentation accuracy [10], [23].

Despite these advancements, existing systems fall short in providing an end-to-end solution for player-specific cricket video segmentation. Previous approaches often focus on event-driven highlights or require manual annotations, limiting scalability for full-match analysis [1], [3], [4]. For instance, while Deanwar's SmartCrick system automates highlight detection, it does not integrate scoreboard OCR and commentary STT for player tagging [4]. Commercial tools like Veo's Player Spotlight offer automated tracking in other sports but are not tailored for cricket's unique context, such as dynamic scoreboards and commentary-driven player mentions [14]. This project builds on these foundations by proposing a novel multimodal framework that combines OCR for scoreboard analysis, STT with NLP for commentary processing, and video-based player detection using models like YOLOv8, MediaPipe, and SlowFast [24], [25], [26]. Unlike prior work, this approach emphasizes automated, scalable segmentation without manual tagging, synchronizing multimodal data to enable real-time filtering of player-specific segments, thereby addressing the core problem of inefficient access to favorite player moments in cricket videos [1], [4], [10].

1.2. Research Gap

The primary challenge in cricket video analysis is that fans, coaches, and analysts must manually navigate hours-long match footage to find moments featuring specific players, which is time-consuming and inefficient. Existing research has made progress in automating cricket video analysis, but it largely focuses on detecting events like wickets or boundaries, often using deep learning models to identify key moments. However, these approaches do not address the need to automatically segment videos based on individual player involvement, leaving users without a streamlined way to access their favorite players' performances.

Current systems fall short in several ways. Most solutions target short clips or highlight reels, lacking the ability to process full-length matches. They often rely on manual tagging or predefined event boundaries, which limits their scalability and automation. Additionally, while some frameworks use audio cues like crowd excitement or visual

features for highlight generation, they do not integrate data from scoreboards or commentary to accurately tag players, missing the context needed for player-specific segmentation.

This project aims to bridge these gaps by developing a novel system that automatically segments cricket videos by player identity. It will combine three key components: extracting player names from scoreboards using text recognition, analyzing commentary audio to detect player mentions, and tracking players in video frames. By synchronizing these data sources, the system will create precise, player-specific video segments without manual intervention. This approach will enable scalable, real-time filtering of player moments, allowing users to instantly access their favorite players' contributions across entire matches, significantly enhancing the efficiency of performance analysis and fan engagement.

1.3. Research Problem

Cricket videos, often spanning hours, require manual navigation to locate segments featuring specific players' involvement in batting or bowling, leading to inefficiency for fans, coaches, and analysts seeking targeted performance insights. Despite advancements in deep learning, no automated system currently exists that can accurately segment these videos by player identity using multimodal data integration, such as Optical Character Recognition (OCR) from dynamic scoreboards to extract player names, Speech-to-Text (STT) from commentary audio to identify player mentions via Natural Language Processing (NLP), and video stream analysis for synchronization and tagging. This gap results in time-consuming frame-by-frame reviews, limiting scalable access to player-centric footage and hindering strategic analysis, coaching, and viewer engagement in a sport with diverse match formats and growing demands for personalized analytics.

2. OBJECTIVES

2.1. Main Objective

This research aims to develop an automated deep learning-based system that can segment cricket videos based on player identity using scoreboard Optical Character Recognition (OCR) and audio commentary.

2.2. Specific Objectives

- To collect and preprocess a dataset of cricket videos with visible scoreboards and audio commentary, ensuring robust data preparation for multimodal analysis.
- To develop and optimize an OCR model to accurately extract player names and contextual information from dynamic scoreboards under varying lighting and motion conditions.
- To design a speech-to-text (STT) system with Natural Language Processing (NLP) to transcribe commentary audio and effectively identify player mentions in noisy environments.
- To create a multimodal synchronization framework that integrates OCR, STT, and video data to produce precise, temporally aligned player-specific video segments.
- To evaluate the system's accuracy, scalability, and real-time performance using metrics such as segmentation precision and processing speed, refining the models for optimal results.
- To design an interactive user interface that enables fans, coaches, and analysts to filter and access player-specific segments efficiently, enhancing user experience and engagement.

3. METHODOLGY

This research project aims to integrate advanced deep learning and multimodal processing techniques to develop an automated system for cricket video segmentation by player identity, addressing the challenge of manual navigation through lengthy match footage. The methodology is structured around key components: video and audio data analysis, deep learning model development, multimodal synchronization, and user interface design, all working synergistically to achieve the research objectives.

- **Video and Audio Data Analysis**
  Central to the system is the processing of cricket video frames and audio commentary to identify players based solely on scoreboard text and commentary audio. A diverse dataset of cricket videos with visible scoreboards and accompanying audio will be collected from various match sources. The methodology involves extracting key frames for Optical Character Recognition (OCR) to identify player names from scoreboards, while Speech-to-Text (STT) processing will transcribe commentary audio to detect player mentions using Natural Language Processing (NLP). This analysis will form the foundation for segmenting player-specific moments in real-time.

- **Deep Learning Model Development**
  The development will focus on designing and training deep learning models tailored for text and audio-based player identification. OCR models such as PaddleOCR and EasyOCR will be optimized to extract player names from dynamic scoreboards, handling variations in lighting and motion. Simultaneously, STT models like OpenAI Whisper or Wav2Vec2 will be trained to transcribe commentary audio accurately, with NLP techniques (e.g., spaCy) applied to recognize player mentions. The training process will utilize annotated datasets of scoreboard images and commentary transcripts, with hyperparameter tuning to enhance name extraction precision and audio recognition in noisy environments.

- **Multimodal Synchronization**

  Synchronization will be achieved by integrating OCR-extracted player names and STT-processed commentary data using a transformer-based or LSTM-based framework. This module will align temporal data from scoreboards and audio to produce precise player-specific video segments, ensuring robust performance across diverse match conditions. The synchronization process will be refined through iterative testing to optimize accuracy and reduce latency.

- **User Interface and System Evaluation**

  The system will feature a user-friendly interface allowing users to upload cricket videos and interactively filter player-specific segments based on automated segmentation results derived from scoreboard and commentary data. Automated data preprocessing will handle video and audio inputs, while user inputs (e.g., preferred player names) will enhance personalization. The system's performance will be evaluated using metrics such as segmentation accuracy, processing speed, and user satisfaction, with adjustments made to improve scalability and real-time functionality.

3.1. System Architecture

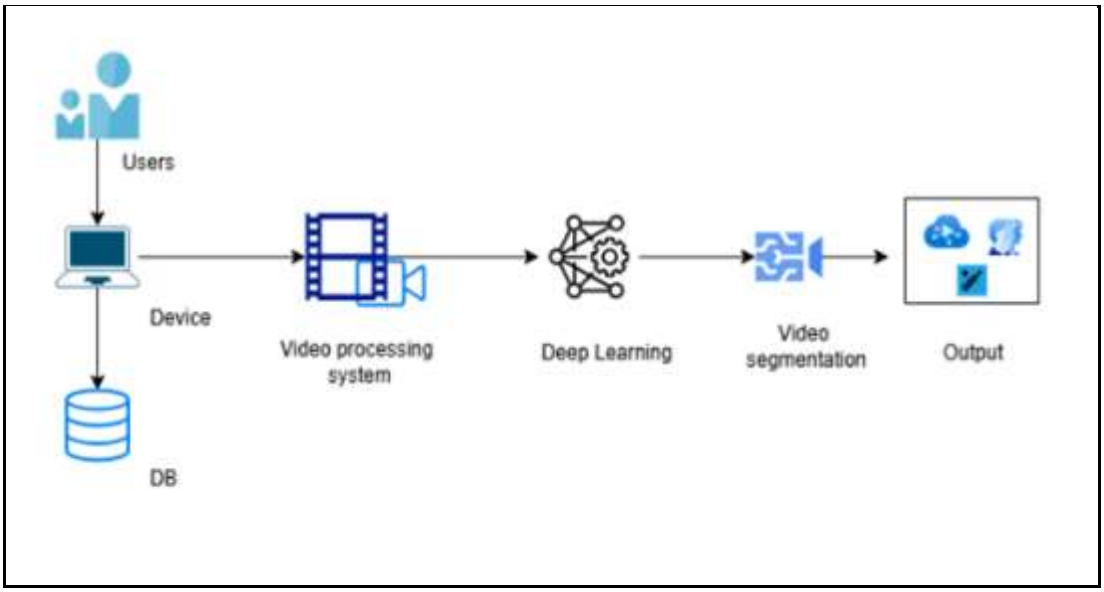

Figure 3-1-level architecture diagram for the proposed system

The above Figure 3.1 shows the overall high-level architecture of the proposed Cricket Video Segmentation System for player identity. Users upload cricket video files using a device that captures and sends the data for analysis. The videos go through a Video and Audio Preprocessing System to standardize and improve quality. This process includes frame extraction and audio normalization before the actual analysis.

Next, the preprocessed data enters a Multimodal Analysis System. Here, Optical Character Recognition (OCR) captures player names from scoreboards. Speech-to-Text (STT) combined with Natural Language Processing (NLP) identifies player mentions in the commentary audio. After identifying players, the system uses a Synchronization Module to match the OCR and STT data with video segments.

Finally, the User Interface and Filtering component produces interactive, player-specific video segments based on the detected player identities. This feature makes it easier for fans, coaches, and analysts to access the relevant content.

Technologies	Python, TensorFlow, OpenCV, Jupyter Notebook, Flask Server,spaCy
Techniques	Video and Audio Preprocessing, Text Extraction, Speech-to-Text Processing, Temporal Synchronization, Model Optimization
Architectures	YOLOv8 ,Transformers,LSTM (Long Short-Term Memory)
Algorithms	Optical Character Recognition (OCR), Speech-to-Text (STT), Natural Language Processing (NLP), Multimodal Fusion

Table 3.1-Technologies

The architectural design includes a Video Upload and Processing component, which lets users upload cricket video files from a device. The system processes these videos using a Video and Audio Preprocessing module to standardize frames and normalize audio. This prepares the data for analysis. The Multimodal Analysis System then uses Optical

Character Recognition (OCR) to extract player names from scoreboards and Speech-to-Text (STT) with Natural Language Processing (NLP) to identify player mentions from commentary audio. It employs techniques like YOLOv8, Transformers, and LSTM for better performance. This analysis provides a thorough identification of player involvement.

The outputs from the OCR and STT/NLP analysis come together in the Synchronization Module. This module matches the extracted player names and commentary mentions with video segments based on time. It ensures precise segmentation based on user-specific factors, such as preferred player inputs, to deliver a customized view of player-centric footage.

This integrated information is used to create interactive, player-specific video segments. The User Interface and Filtering component turns the segmented data into useful outputs. Users can filter and access moments featuring specific players through a simple interface. These segments are displayed as real-time, customizable views, making them relevant and practical for fans, coaches, and analysts based on their input.

The system's workflow involves a steady flow of data from uploaded cricket videos to analysis components. This ensures that player identification and segmentation are timely and relevant. Users engage with the system through an easy interface, getting feedback about player-specific segments, along with the option to refine their queries, which continuously improves based on their preferences.

Overall, the proposed architecture meets all project goals by processing uploaded cricket videos with a focus on personalized segmentation. It learns from user feedback to create a dynamic and responsive system. This approach improves users' ability to manage and analyze player performance effectively by integrating both scoreboard and commentary data.

3.2. Software Solution

The Software Development Life Cycle (SDLC) aims to cut the cost and time of developing the ECG-Based Disease Detection Module while improving the quality of the final product. It does this by following a clear plan that identifies and tackles common issues in software development projects. The process begins by evaluating existing cardiovascular monitoring systems for possible weaknesses. This allows for a more efficient and effective development strategy. After identifying gaps in current systems, the SDLC lays out the specific needs for the module. This is followed by the phases of analysis, planning, design, development, testing, and deployment. The SDLC works to prevent expensive mistakes, such as overlooking feedback from users or medical professionals, by using a systematic approach that minimizes extra work and fixes needed after deployment. The testing phase is an essential part of the SDLC, ensuring code quality at every step. By focusing on testing, the project can save time, resources, and effort in the long run. The following image outlines the stages of the SDLC.

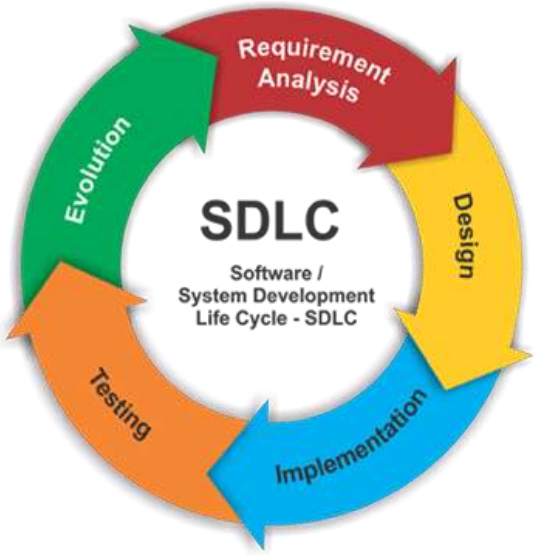

Figure 3-2-Stages of Software Development life cycle

• Requirements gathering and analysis

In the initial phase of the project, we will gather all requirements related to cricket video segmentation by player identity for the Cricket Video Segmentation System. We will work with cricket analysts, coaches, and technology specialists to ensure the system meets both technical and practical needs. Guidance from experts in sports video analysis and deep learning will help us develop methods for accurate and effective player identification. This will make sure the system's features meet user needs and deliver practical outcomes.

• Sensing Information and Data

To make sure the system works well, we will take a thorough approach to data collection. Users will upload cricket video files, which will serve as the basis for identifying player involvement. We will process these videos using Optical Character Recognition (OCR) to extract player names from scoreboards. We will also use Speech-to-Text (STT) with Natural Language Processing (NLP) to detect player mentions from commentary audio. Data integration will combine these insights with user-provided preferences, like specific player names, offering a customized segmentation experience.

• Engaging Stakeholders

Potential users, including cricket fans, coaches, analysts, and technology experts, are essential in shaping the system. We will collect insights into their needs and expectations regarding features such as accurate player identification, interactive filtering, and ease of use through surveys and focus groups. Based on this stakeholder input and the expertise of cricket analysts, the system will be designed to be usable and effective, meeting the preferences and requirements of its target audience.

Feasibility Study (Planning)

• Economic Feasibility

The economic feasibility study looks at the project's potential benefits compared to development costs. Without a solid financial plan, the project could fail. To ensure the proposed Cricket Video Segmentation System is both affordable and effective, we will prioritize a cost-effective implementation strategy that balances performance with resource use, such as using open-source tools.

• Scheduled Feasibility

A schedule feasibility analysis reviews anticipated project timelines, which is crucial for success. Not completing the project on time may hinder achieving its goals. As shown in Figure 4.1 (Gantt chart), sticking to the proposed timeline for each task will help keep progress steady and ensure completion within the designated period.

• Technical Feasibility

Technical feasibility planning focuses on evaluating the skills and expertise needed to develop the system, such as knowledge of deep learning frameworks (e.g., TensorFlow), video/audio processing tools (e.g., OpenCV), and software architectures. This analysis will help determine if the necessary infrastructure, technology, and skills are available or can be obtained. By assessing these factors thoroughly, the project team will ensure the system is developed efficiently and meets technical standards.

Design (System and Software Design Documents)

After the planning phase, we will create system and software design documents to support the development of the overall system diagram**.**

Sequence Diagram

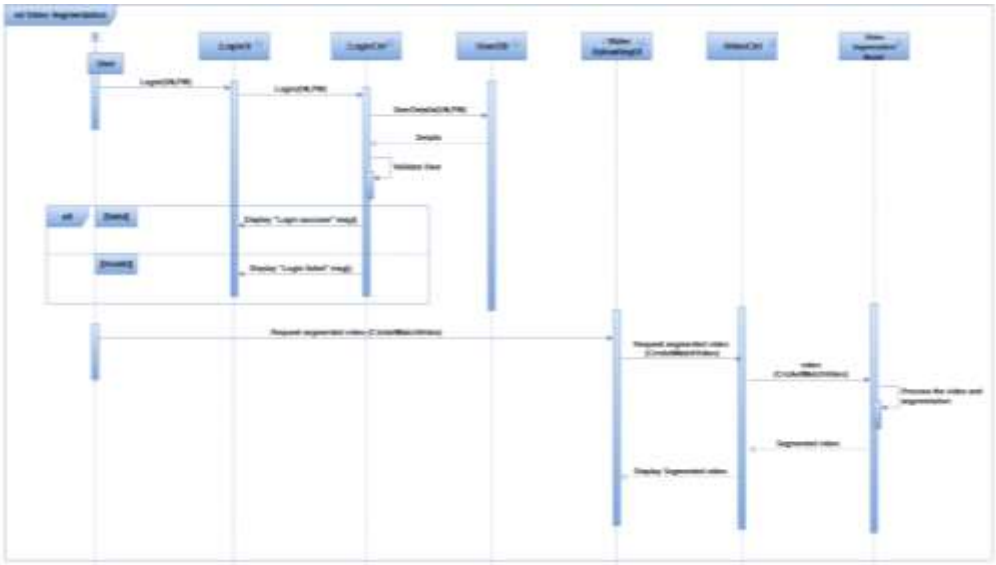

Figure 3-3-Sequence diagram

Use Case Diagram

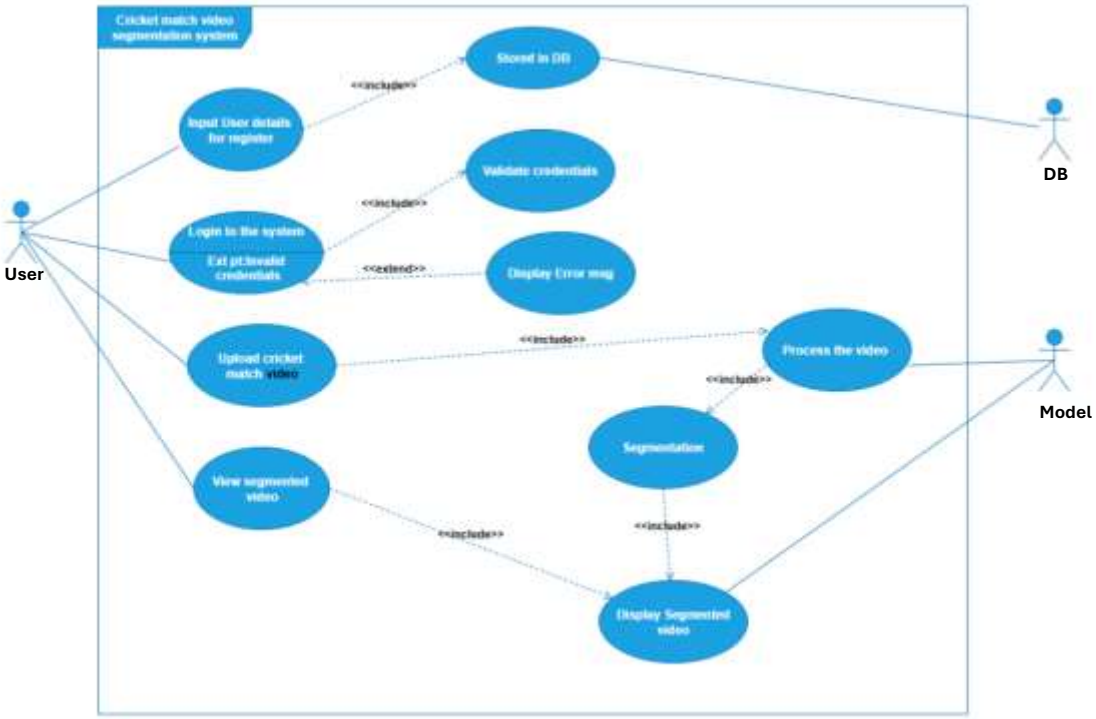

Figure 3-4-Use case diagram

Testing (Track and Monitor)

During the software testing phase, several processes will take place, including unit, component, integration, system, and user acceptance testing. These processes will help identify gaps, missing requirements, errors, and bugs in the Automatic Cricket Video Segmentation by Player Identity Module. The tests will ensure the quality of the software by addressing issues and confirming its reliability and accuracy in segmenting cricket videos based on player involvement.

3.3. Commercialization

Currently, in cricket video analysis, viewers, coaches, and analysts often have to manually sift through lengthy match recordings to pinpoint when a specific player was active. This manual approach is not only inefficient but also takes up a lot of time and effort, especially with full-length matches that can last for hours. This process slows down training sessions, makes performance reviews more difficult, and delays the creation of highlights for individual players.

The proposed system aims to solve this problem by automatically segmenting the video content based on player involvement. It does this by using scoreboard optical character recognition (OCR), speech-to-text commentary analysis, and advanced deep learning models. By using these methods, the system can effectively identify and isolate the parts of the match where a certain player is batting or bowling.

As a result, instead of spending hours looking through entire matches, users will be able to quickly access the specific times and segments that relate to each player's activities. The solution is designed to be fast, accurate, and easy to use, ensuring that even those with limited technical skills can take advantage of it. Moreover, this system can be marketed well to a broad audience, including cricket coaches, professional analysts, sports broadcasters, and fans who want to revisit their favorite player's performances. Overall, the system simplifies post-match analysis, player performance evaluation, and highlight retrieval, making them much more efficient.

Future Scope

The future of the Automatic Cricket Video Segmentation Module includes adding auto-navigation features. This will allow the system to directly take users to the area of play for a specific player when searched by name. Advanced filtering options, such as batsman-wise, bowler-wise, or role-based segmentation, will be added to make video retrieval more focused and efficient. The system will also have a recommendation feature that suggests videos based on users' search habits, improving personalization and engagement. In addition, future updates will emphasize using advanced multimodal methods to enhance the accuracy of OCR and commentary alignment, especially in noisy or unclear recordings. These improvements will help the system become a more precise, user-friendly, and intelligent cricket analytics tool for analysts, coaches, players, and fans.

4. PROJECT REQUIREMENTS

4.1. Functional Requirements

1. The system should analyze uploaded cricket match videos and provide segmented videos.

2. The system should detect player identities by using Optical Character Recognition (OCR) on scoreboard frames and speech-to-text analysis of commentary audio.

3. The system should highlight and tag video frames or segments with player-specific information. It should ensure that this information is clear through visual markers or timestamps.

4.2. Non-Functional Requirements

1. User-Friendliness-The system should have a web interface that is easy to use, responsive, and visually attractive. It should simplify video uploads and segment retrieval for non-experts across devices, including desktops and laptops.

2. Reliability- The system should run smoothly without interruptions. It must ensure accurate player identification and segmentation. All user-uploaded data should be encrypted and protected to meet privacy standards.

3. Performance- The system should provide fast and accurate segmentation of cricket videos. It should deliver results within seconds to minutes, depending on video length, and handle high user demand effectively.

4. Availability-The application should be open to all users, no matter the language. It must offer multilingual support and be available 24/7 for timely analysis.

5. Maintainability-The system should be easy to maintain. It needs comprehensive documentation for the segmentation module, a modular design, and a simple process for updating features or fixing bugs.

4.3. System Requirements

The system requirements for the Player-Based Segmentation Module include strong hardware and software to efficiently process cricket videos and manage user data. The hardware should have a powerful CPU or GPU for deep learning tasks and reliable network connectivity for smooth video uploads and web access. Development frameworks need to include TensorFlow and OpenCV for analyzing video frames and detecting players, along with Streamlit for backend operations and Node.js for server connectivity. A database system is essential for managing user-uploaded videos, segmentation results, and metadata. The system must provide quick processing, ensure data security through encryption, and support secure API integration. It should also be able to handle increasing video volumes and user numbers while maintaining high reliability and fault tolerance.

4.4. User Requirements

When designing the web interface for cricket fans, coaches, and analysts who want player-specific video analysis, it is important to consider their needs. The interface should be interactive and simple. It should use clear language to help users upload cricket videos and understand segmented results, avoiding technical terms or complicated instructions. The system should be engaging and easy to use, encouraging regular use for performance analysis while creating a smooth experience for non-expert users.

Expected Use Cases

Use Case: Upload and Segment Cricket Video

• **Actor:** Cricket Fan, Coach, or Analyst

• **Description:** The user uploads a cricket match video to the system for segmentation based on player identity.

• **Preconditions:** The user has access to a compatible device, such as a computer or laptop, and a valid cricket video file.

• **Trigger:** The user navigates to the web interface and selects the upload option.

• **Basic Flow:**

1. The user logs into the system if required.

2. The user uploads a cricket video via the interface.

3. The system processes the video using deep learning models for player detection, OCR for scoreboard analysis, and speech-to-text for commentary.

4. The system segments the video by player identity and tags segments with timestamps.

5. The system displays the segmented results.

• **Postconditions:** The user receives a report with player-specific video segments and metadata.

• **Exceptions:** An invalid file format triggers an error message; a processing failure prompts a retry option.

Test Case 1: Segmentation Accuracy

• **Objective:** Evaluate the system's ability to accurately segment cricket videos by player identity.

• **Method:**

- **Dataset:** Use diverse cricket match videos with different player activities, such as batting and bowling, and various conditions like lighting and camera angles.
- **Metric:** Calculate Precision, Recall, F1 Score, and User Satisfaction Rate based on segmentation accuracy and relevance.
- **Test:**
  - **H0:** The segmentation accuracy (Precision, Recall, F1 Score) of the model-generated player segments is comparable to manual segmentation methods.

    - **H1:** The segmentation accuracy (Precision, Recall, F1 Score) of the model-generated player segments is significantly higher than manual segmentation methods.
  - **Statistical Test:** Perform paired t-tests or Mann-Whitney U tests to compare performance metrics.

5. PERSONAL AND FACILITIES

- Facilitators
    - Prof.Anuradha Karunasena – Sri Lanka Institute of Information Technology
    - (SLIIT).
    - Ms. Adya Dissanayake - Sri Lanka Institute of Information Technology (SLIIT).
- Facilities
    - ESPNcricinfo Website
    - YouTube Cricket Matches
    - Other Cricket Match Video Providing Websites

6. GANTT CHART

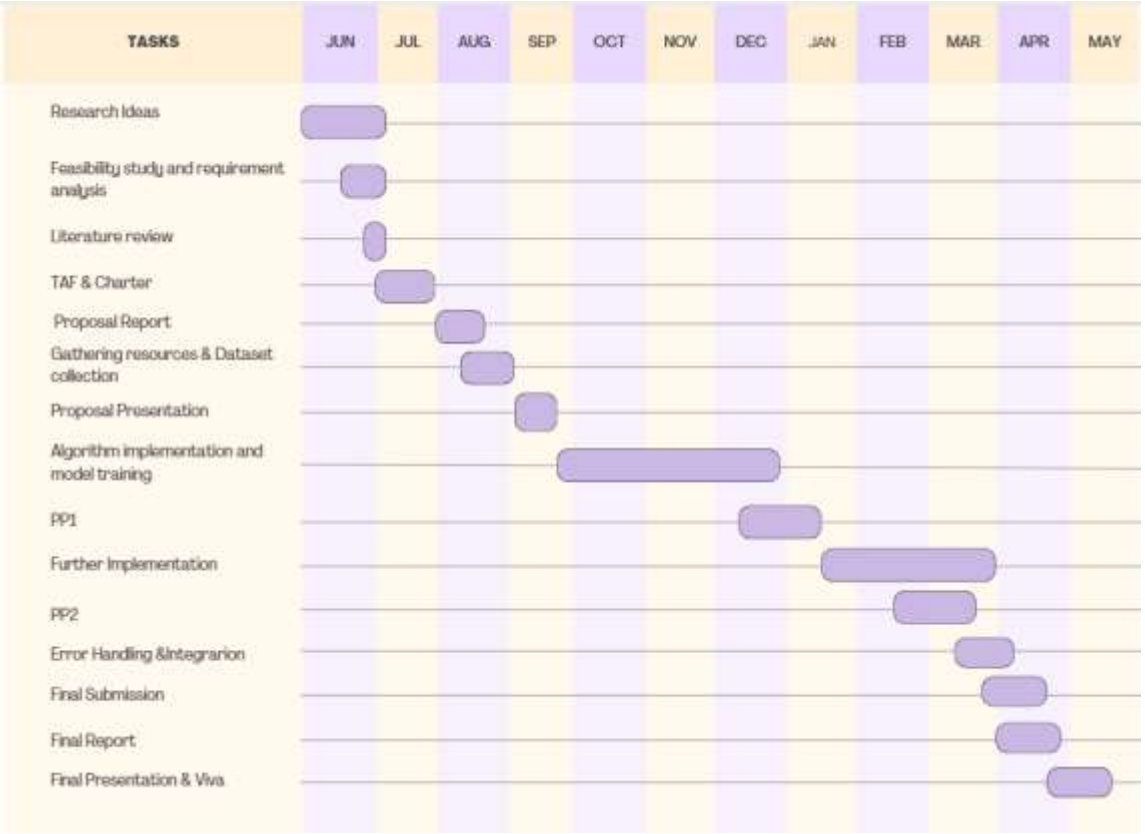

Figure 6-1-Gantt Chart

7. WORK BREAKDOWN STRUCTURE (WBS)

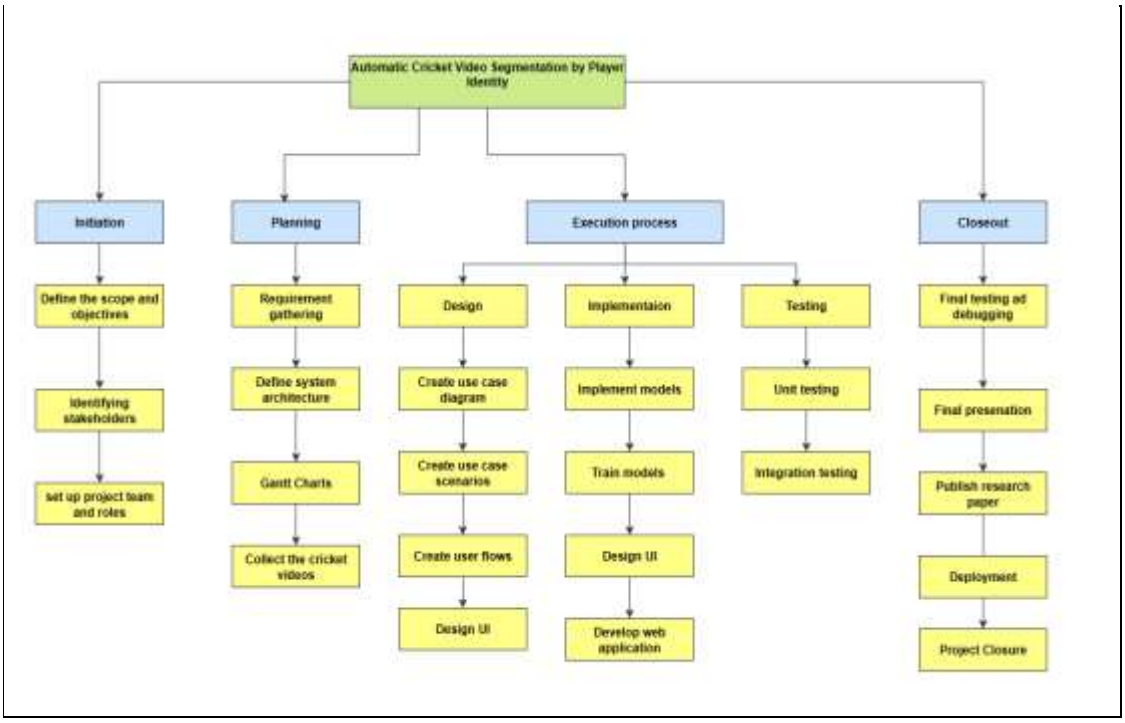

Figure 7-1-WBS

8. WIREFRAMES

Figure 8-1-Wireframe 1

Figure 8-2-Wireframe 2

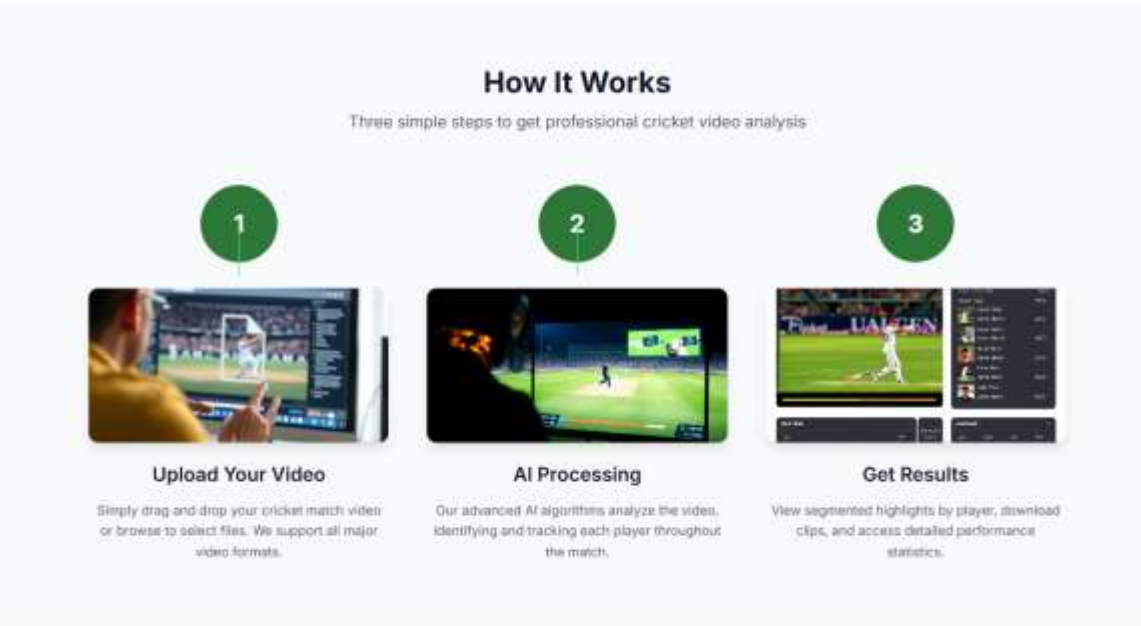

Figure 8-3-Wireframe 3

Figure 8-4-Wireframe 4

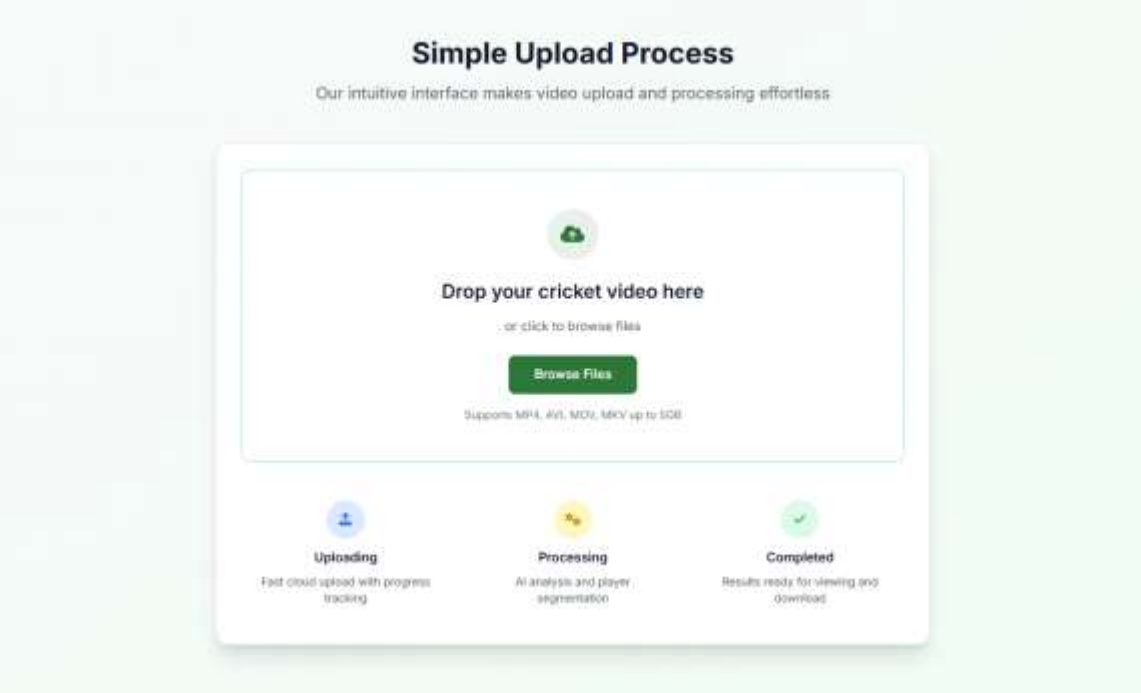

Figure 8-5-Wireframe 5

Figure 8-6-Wireframe 6

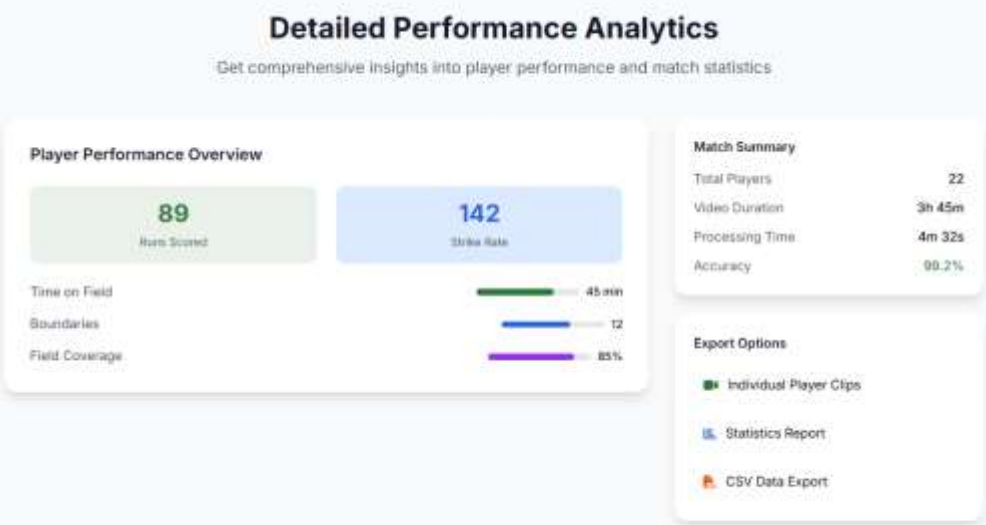

Figure 8-7-Wireframe 7

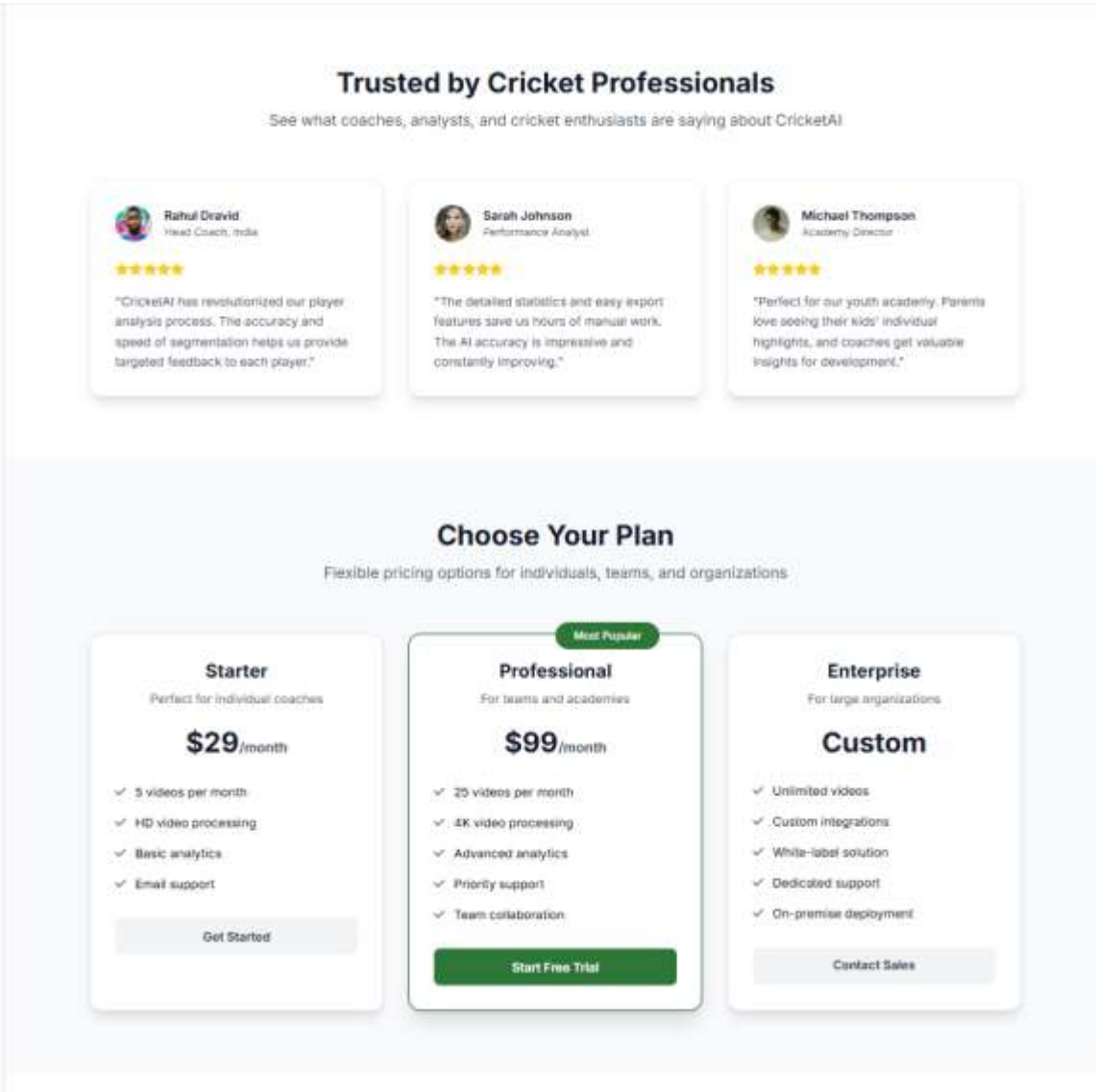

Figure 8-8-Wireframe 8

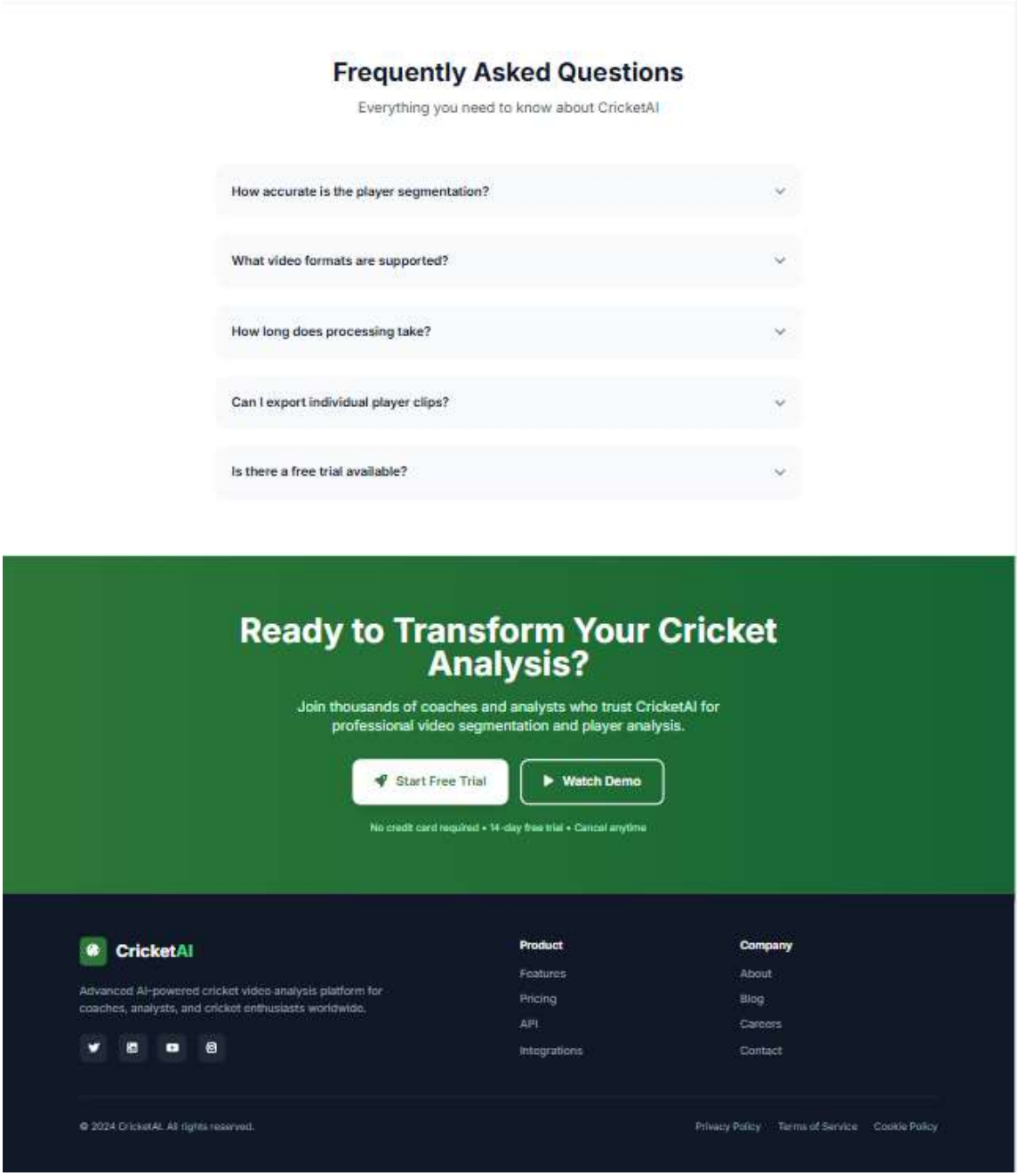

Figure 8-9-Wireframe 9

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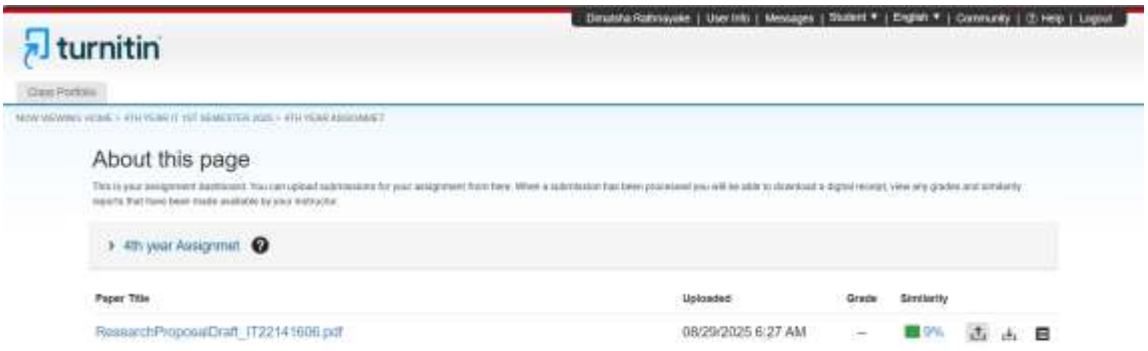
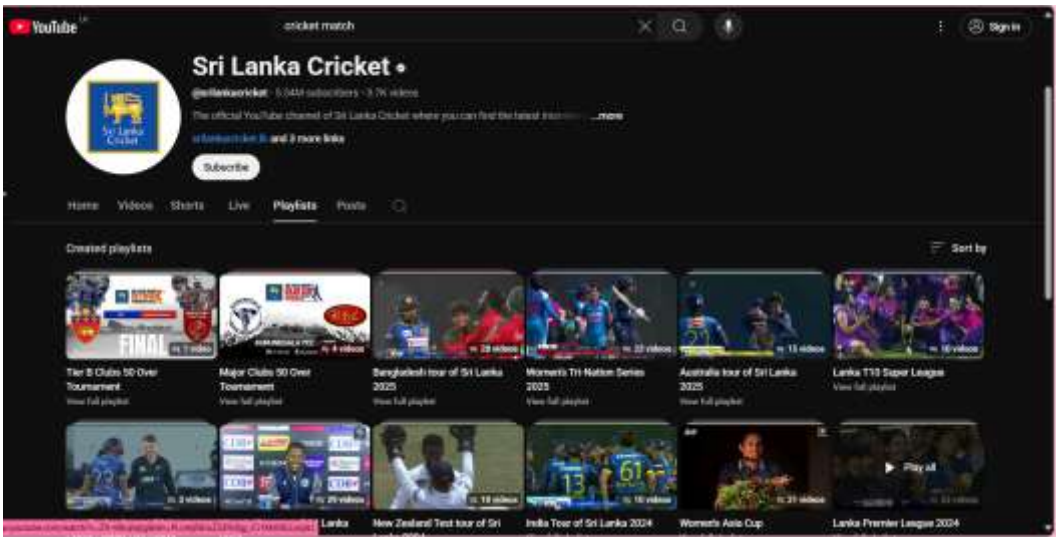
APPENDICES

Appendix A: Plagiarism



Appendix A.1:Plagiarism

Appendix B: Sample Datasets



Appendix B.1: Data set